Project Phase 1: Requirement Engineering and Design

CMPT 276: Introduction to Software Engineering
Simon Fraser University

Group 1

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1 Overall Plan and Game Description

Game Entities & Core Mechanics

- **The Thief (Main Character):** Controlled via keyboard (up/down/left/right). The Thief starts at the "Entry Point" (a cell on the boundary wall) and can move a maximum of one cell per tick.
- **Cash Stacks (Regular Rewards):** Multiple stacks of cash are scattered across the board. The Thief must collect all of them before the "Getaway Van" (exit cell) becomes active for a win.
- **Guards (Moving Enemies):** Security guards that patrol the map. They move exactly one cell per tick, automatically choosing a direction that minimizes their distance to the Thief's current position. If a Guard walks over a Cash Stack or Security Hazard, it does not affect or trigger the item.
- **Security Hazards (Punishments):** Stationary "Silent Alarms" or "Traps" placed on specific cells. Stepping on these deducts a set amount from the player's total score, and then the trap is removed.
- **The Rare Diamond (Bonus Reward):** A high-value diamond appears randomly on the board for a limited number of ticks before disappearing. While not required to win, it significantly boosts the final score.

Winning & Losing Conditions

- **Victory:** The player wins by collecting every "Regular Reward" cash stack on the map and successfully reaching the "Exit" cell (the Escape Van) located on the boundary wall.
- **Defeat:** The game ends immediately if a Guard moves into the Thief's cell or if the Thief moves into a Guard's cell. Additionally, the player loses if their total score drops below zero due to hitting too many Security Hazards.

The Game Environment

When the game starts, an initial map is loaded to populate the walls, characters, and items. The game is played on a 2D grid surrounded by four main boundary walls, with two notable openings for the entry (player starting area) and the escape van. Internal walls and barriers create a maze-like environment that restricts movement for both the player and the guards. The screen will display the player's current score and a timer showing the elapsed playtime.

Technical Architecture & Grid Logic

The game operates on a discrete 2D coordinate system where the board is represented as a `Cell[x][y]` array. Instead of complex pixel-overlap detection, "collisions" are handled via coordinate matching during a game "tick". Each cell maintains a state (`WALL`, `EMPTY`, `TRAP`, `REWARD`, `EXIT`). Only one moving entity can occupy a single coordinate at any given time. When the Thief moves to a cell, the engine checks the state. If the state is a reward or trap, the logic triggers an immediate score update and reverts the cell to `EMPTY`.

2 User-Interface Mockups

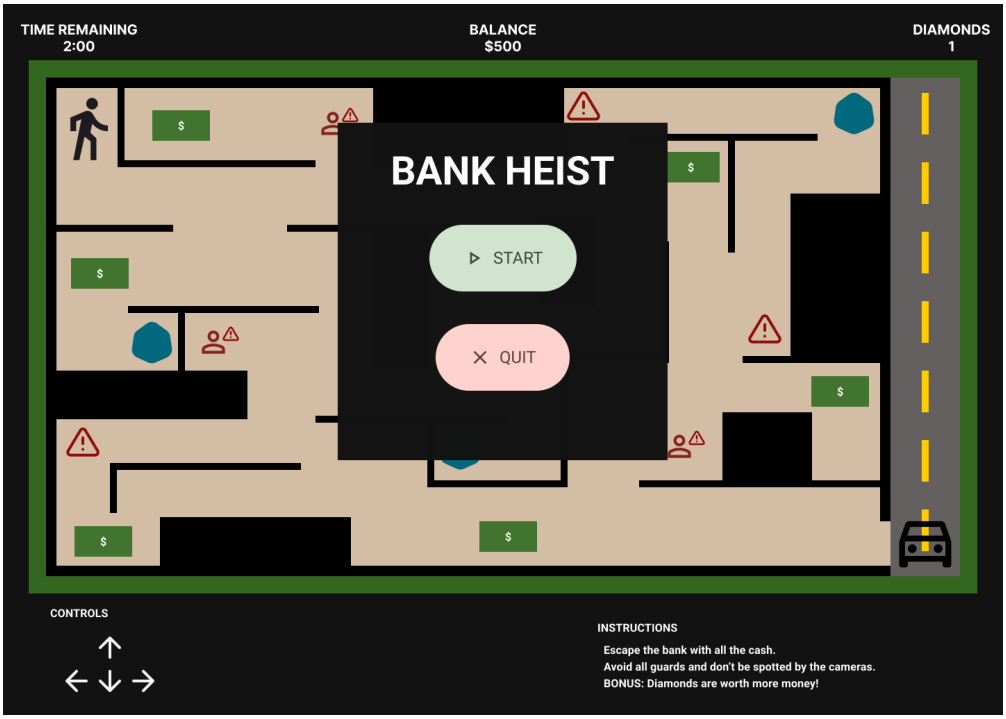


Figure 1: Main Menu Mockup

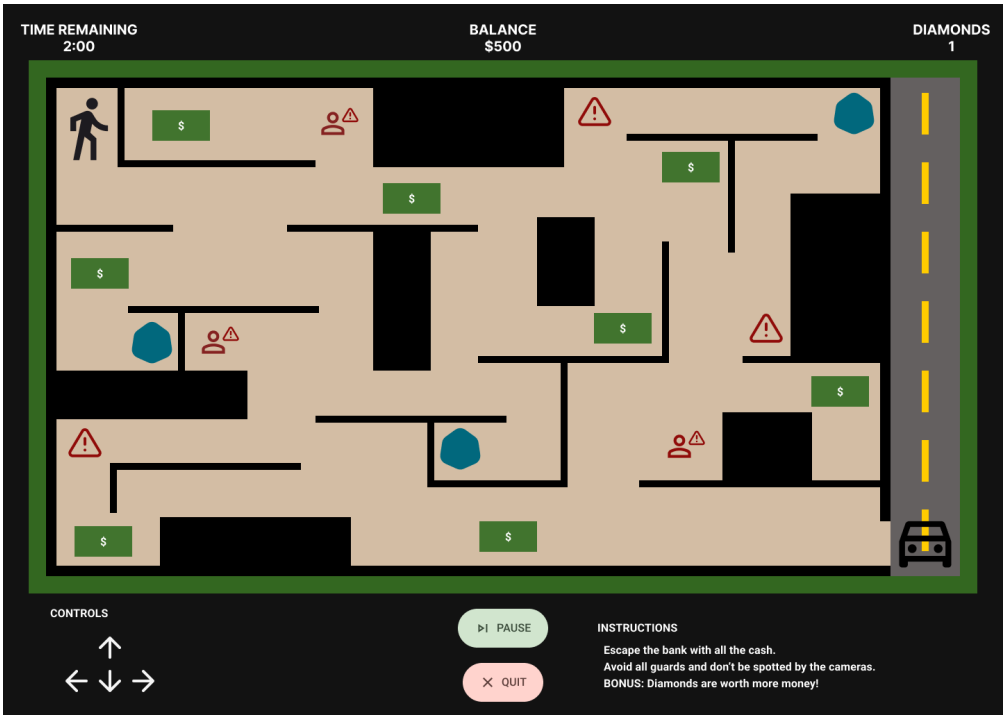


Figure 2: Main Game Screen Mockup

3 UML Class Diagrams

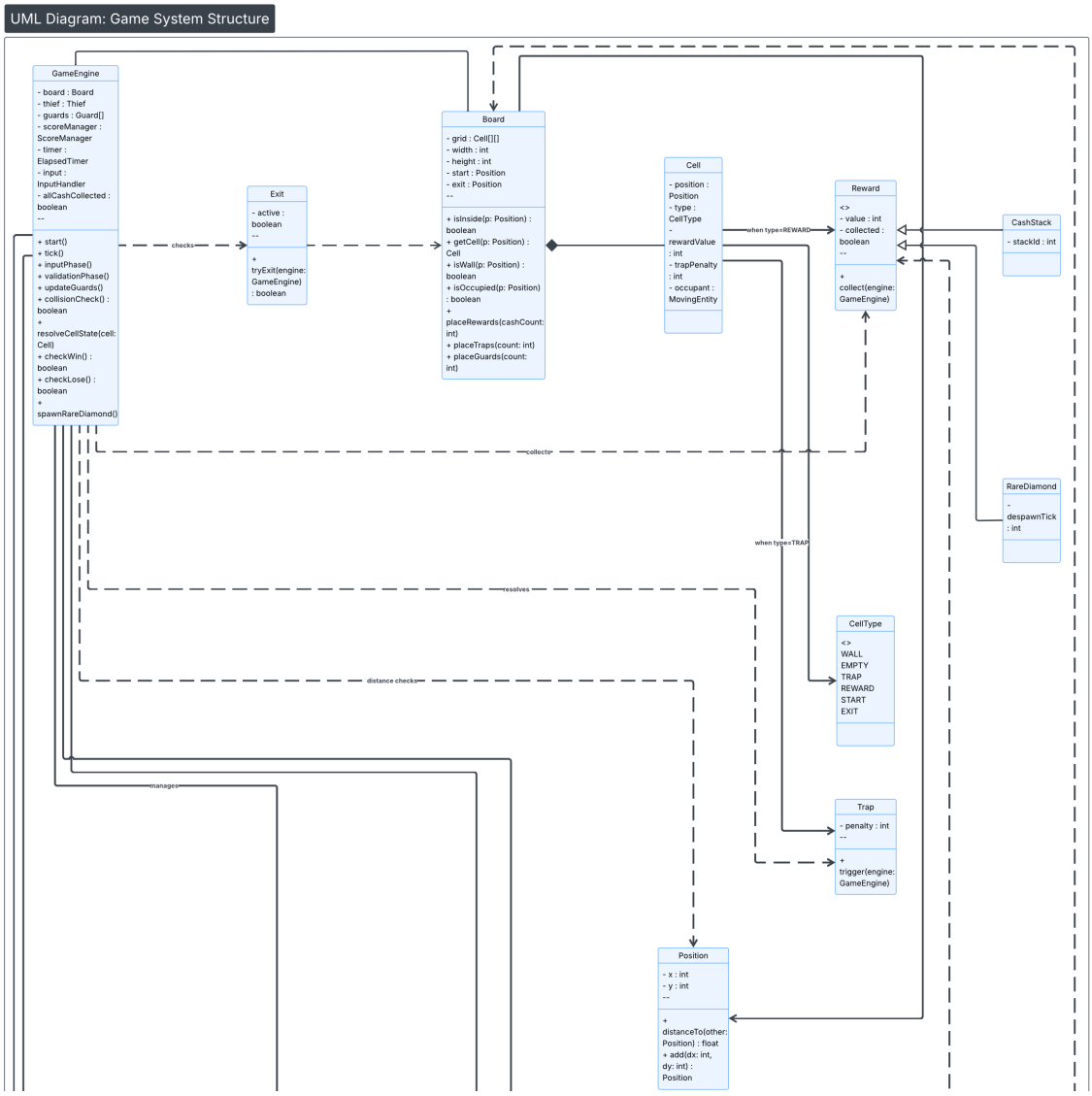


Figure 3: System UML Class Diagram - Part 1

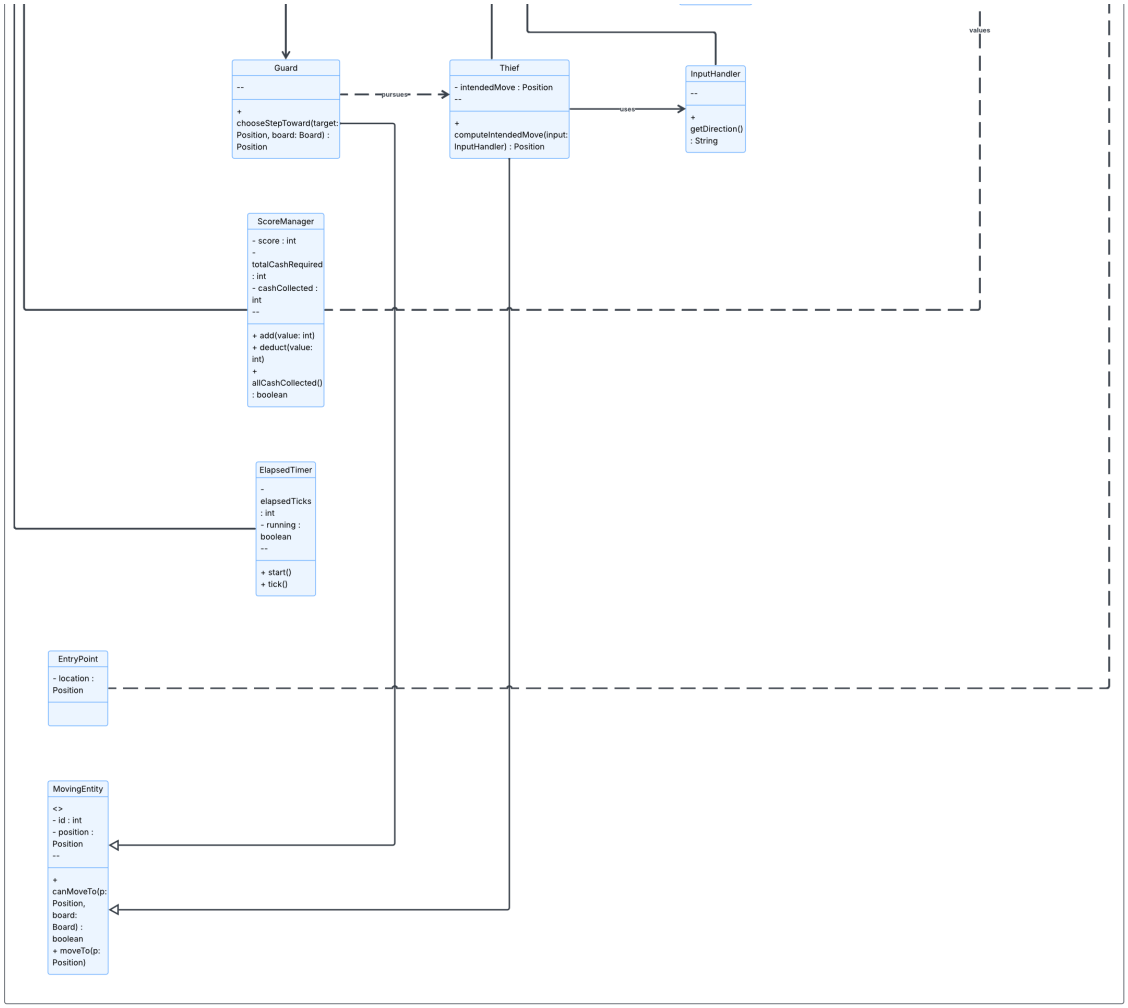


Figure 4: System UML Class Diagram - Part 2

4 Team Contract

Communication

- Discord will be used for all text communication and online meetings.
- In-person meetings will happen whenever necessary.
- Our expectation is that everyone will check Discord messages daily to ensure any issues are promptly addressed.

Meetings

- We will meet at least once per week (usually Sunday @ 7:00pm) to touch base.
- We will schedule extra meetings whenever required.
- Meetings will either be in-person on campus or online via a Discord call.

Tasks

- Everyone will have an opportunity to volunteer for tasks based on what they are interested in and/or what they are good at.
- Remaining tasks will be assigned by group consensus to balance the workload.
- We aim to split complex tasks (i.e., game logic, UI) evenly so everyone contributes to the code.

Deadlines

- We will set internal deadlines 2 days before the official course due date.
- This buffer time will allow for everyone to review the work completed.
- There will be check-ins to ensure tasks are not left until the last minute.

Conflict Resolution

- Since we are a group of 5, any disagreements will be resolved by a majority vote.
- If a member misses a deadline, they must notify the group via Discord at least 24 hours in advance. Missed work must be completed within an agreed-upon grace period.
- If a member consistently fails to contribute or attend meetings, the issue will be discussed during a meeting. If unresolved, it will be documented and reflected in the end-of-semester peer reviews.

Signatures

By typing our names below, we agree to the terms of this contract.

1. Brian Chan
2. Keith Ng
3. Daniil Bugakov
4. Mohamad Osman
5. Dev Tomar